

Emerging Technologies and Big Data



Course Description

In this course, you will explore contemporary systems and technologies impacting the field of data analytics, including the cloud, AI, and machine learning. You will also gain exposure to areas of technology that provide opportunities for future professional specialization, such as emerging big data technologies that can support the work of data analysts.

Projects

The course goals are communicated through three competency statements instead of through course outcomes. Competencies represent the knowledge and skills relevant to your field. Additionally, there is not a single final project like you may have seen in other courses. Instead, there are two smaller projects. Project One will assess the first and second competencies. Project Two will assess the second and third competencies. Still, the amount of material covered, the level of difficulty, and the workload expectations are all typical for a 200-level course.

Project One (Module Four Submission)



Create a feasibility study for migrating client data to a cloud environment that will include an explanation of special handling requirements for big data.

In this project, you will demonstrate mastery of the following competencies:

- Explain the concepts of cloud-based architectures
- Differentiate between the functions of big data technologies

Project Two (Module Seven Submission)



Create a proposal to request the use of new technology tools to analyze big data collected as part of a new IoT initiative, and to share your recommendations for how AI and machine learning can benefit the organization.

In this project, you will demonstrate mastery of the following competencies:

- Differentiate between the functions of big data technologies
- Explain how the internet of things and artificial intelligence can support and be used in an organizational environment